LECTURE 5 – REVIEW

Fixed Income

Duration, Convexity & Immunization

Esade MFin 2026 · Presented by · Group 6

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Agenda

1. Duration Review
2. Dollar Duration & the Zero-Investment Portfolio
3. Price Value of a Basis Point
4. Convexity
5. Immunization
6. In Practice: the Butterfly Strategy

Fixed Income: Understanding Duration

Interest Rate Sensitivity

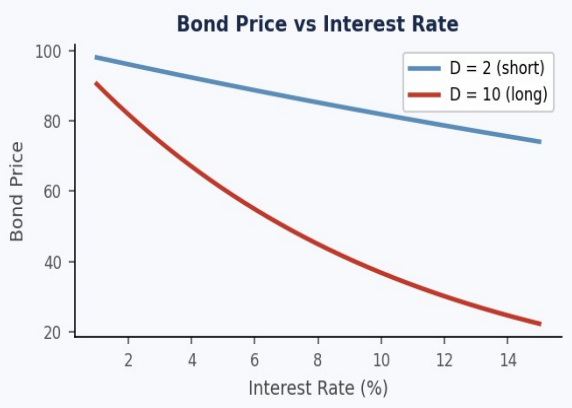
① Why Duration Matters

Bond prices and interest rates move **inversely**. Duration quantifies this price sensitivity.

$$D_P = -\frac{1}{P} \times \frac{dP}{dr}$$

$$dP = -D \times P \times dr$$

***Intuition**: A bond with D = 5 loses 5% in price for every 1% rise in rates.*



② Zero-Coupon Bond

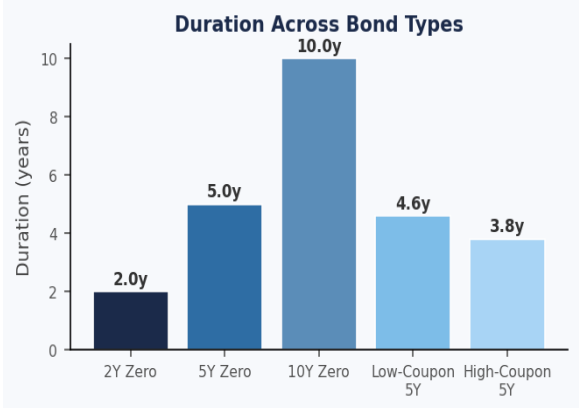
A single cash flow at maturity- all interest rate risk concentrated at one date.

$$P_Z = 100e^{-r(T-t)}$$

$$D_Z = T - t$$

Duration = Time to Maturity

***Intuition:** The further the payment, the greater the price sensitivity to rate changes.*



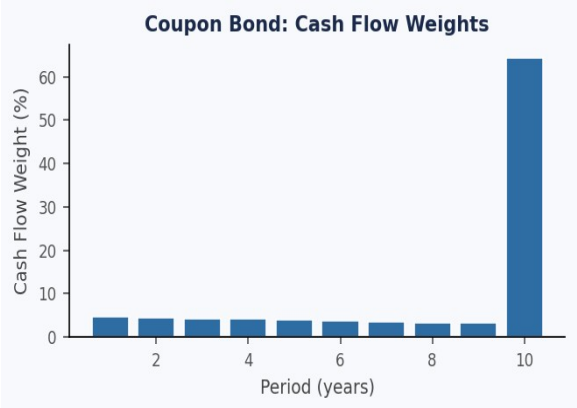
③ Coupon Bond

Think of a coupon bond as a portfolio of zero-coupon bonds. Duration is the PV-weighted average maturity.

$$D_C = \sum_{i=1}^{n} w_i T_i$$

↑ Higher coupon → lower duration
(cash flows arrive sooner)

↑ Higher rates → lower duration
(distant flows discounted more)



④ Key Takeaway

- Duration measures **% price sensitivity** to a 1% rate change.
- Zero-coupon bond: **Duration = Time to Maturity**
- Coupon bond: **Duration = PV-weighted avg. maturity**
- Higher coupons or higher rates → **lower duration**
- When rates ↑, bond prices ↓: and high-duration bonds fall more

Next: Dollar Duration

Duration tells us % sensitivity. To translate into actual monetary gain/loss, we use **Dollar Duration** (DV01).

Fixed Income: Dollar Duration and the Zero-Investment Portfolio

① Dollar Duration

Dollar duration is the **actual euro change** in bond value for a unit move in rates.

$$D_P^{\$} = -\frac{dP}{dr}$$

$$D_P^{\$} = P \times D_P$$

Intuition: *Duration is a percentage — you cannot add % across positions with different market values. D$ converts it into euros.*

Why D$ matters more than D:

Two bonds, same +1 bp shock:

Bond X: D = 5, P = €10M
D$ = 10M × 5 = €50M

Bond Y: D = 3, P = €200M
D$ = 200M × 3 = **€600M**

Lower D, but 12× the real risk exposure!

Takeaway: Duration alone can mislead — position size matters. D$ is what trading desks focus on.

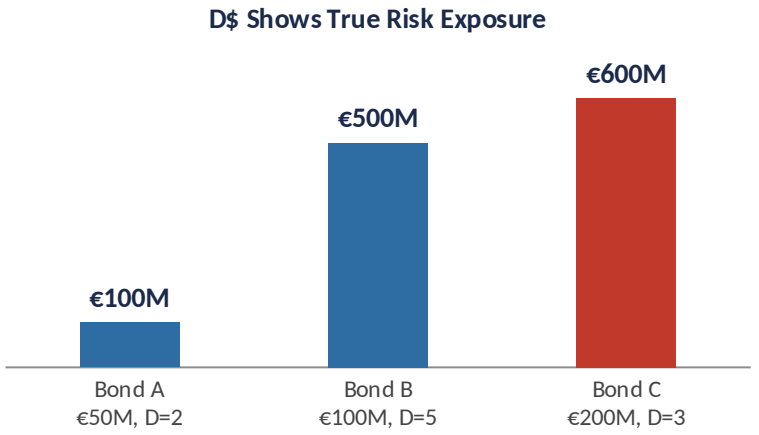
② Portfolio D$ & Zero-Investment

Simply **sum** dollar durations across all positions — no weighting needed.

$$D_W^{\$} = \sum_i N_i \times D_{P_i}^{\$}$$

Zero-Investment Portfolio (W = 0)
Long-short position where net value = 0.
Standard D = Σ $w_i D_i$ is **undefined** (divides by zero).
D$ still works — no division by portfolio value.

Dollar duration avoids the weighting problem entirely



③ Key Takeaway

- D$ = actual euro P&L per unit rate move.
- D$ = P × D for any single bond.
- Portfolio: sum individual D$ across positions.
- Zero-investment: standard D fails; D$ works.
- Dollar duration allows direct comparison across positions.

Next: PVBP (PV01)

Dollar duration gives us a P&L number. Shrink the rate move to one basis point and you get **PVBP** — the most practical risk metric on a trading desk.

PVBP / PV01 - Price Value of a Basis Point

What it measures

It answers the desk's daily question: "**if rates move 1bp, how many € do I make or lose?**" It is the most-used risk metric on any fixed income trading floor. **It's almost like a unit of interest-rate** risk.

PVBP tells a trader exactly how many euros are at stake for every basis point move in rates.

Formula 1 - from Dollar Duration

$$PVBP = -D\$ \times B_p$$

D$ = Price × Modified Duration & 0.0001 = one basis point

Duration → % price sensitivity **PVBP** → € price sensitivity *Same idea, different units.*

Fast, Simple but linear so it doesn't included convexity so only good for small yield changes.

Formula 2 - two-sided repricing (use this in Excel)

$$PVBP = \frac{[P(-1bp) - p(+1bp)]}{2}$$

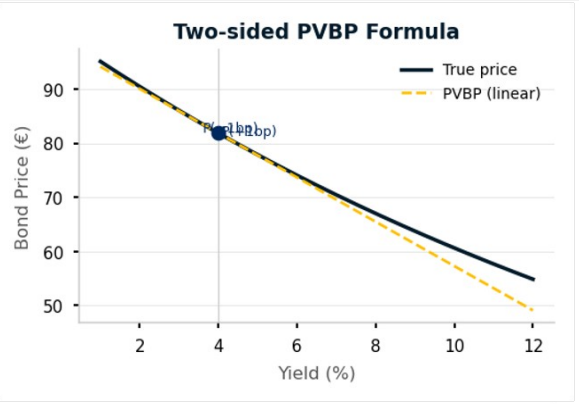
Remember to use : **Dirty price (clean+accrued) & modified duration**.

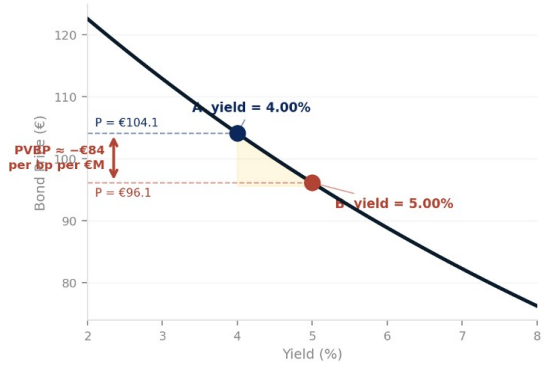
More accurate and preferred. Captures curvature both sides, however this means its computationally heavier, needed two bond valuations.

Portfolio PVBP is additive across positions

$$PVBP_P = \sum N_i \times PVBP_i$$

Sum the dollar exposures across every position, no weighting, no scaling needed.





Key properties

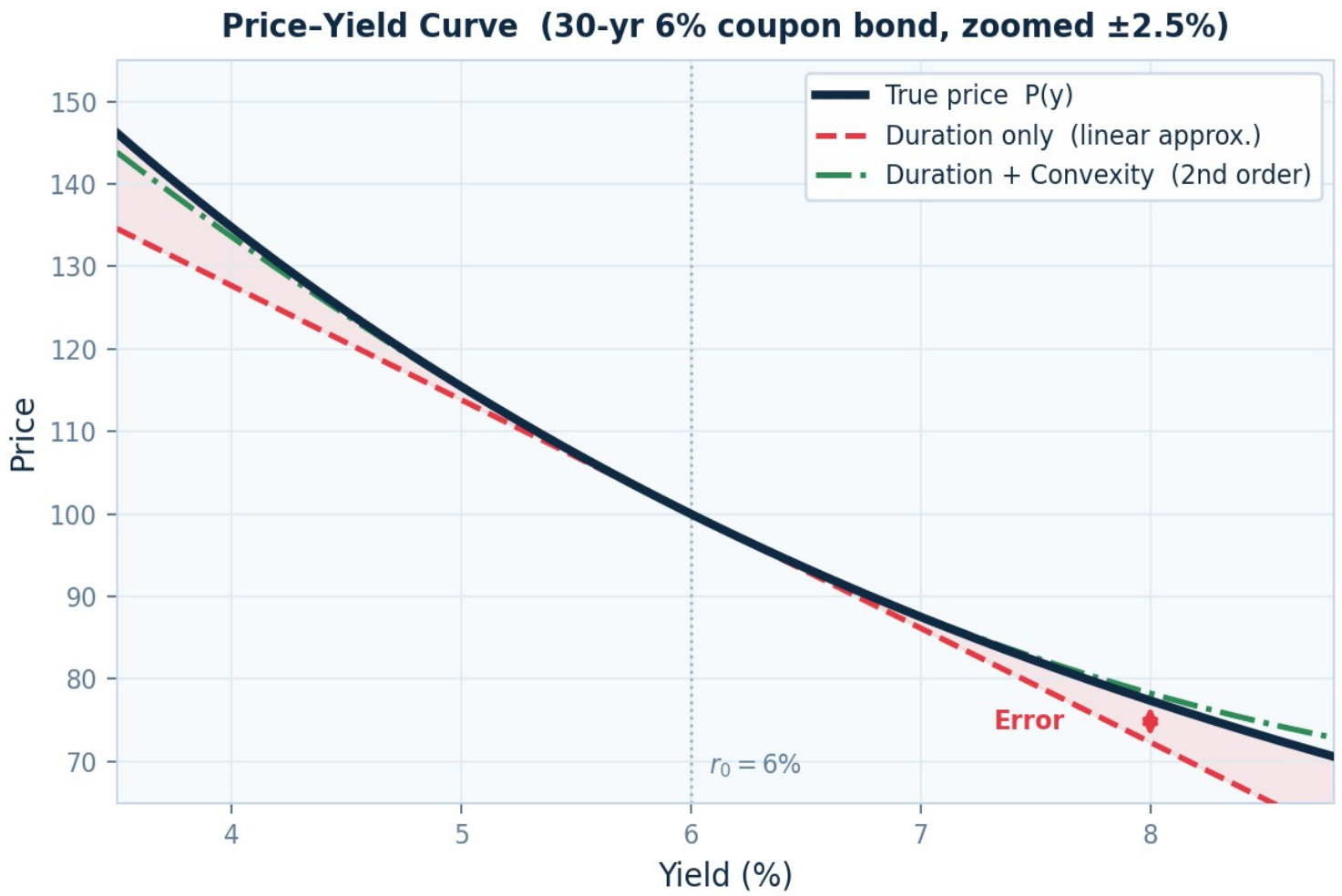
Always negative for long positions
rates ↑, price ↓

Always positive for short positions
Rates ↓, price ↑

Duration Is Linear - Bond Prices Are Not

Price-Yield Curve

Approximating the bond price curve.



30-yr 6% coupon bond · zoomed ±2.5% around r_0 · shaded = duration error

The Derivative Ladder

Slope

$$\frac{dP}{dy}$$

Raw Price Sensitivity

Duration

$$-(\frac{1}{P}) \times \frac{dP}{dy}$$

Standardized Slope

Curvature

$$\frac{d^2P}{dy^2}$$

Rate of Change of Slope

Convexity

$$\frac{1}{P} \times \frac{d^2P}{dy^2}$$

Standardized Curvature

Everyone Likes Convexity

Convexity corrects duration in your favour — whether yields rise or fall.

Convexity defined:

$$C = \frac{1}{P}\frac{d^2P}{dy^2} = \frac{1}{P}\sum_t \frac{t(t+1)\,CF_t}{(1+y)^{t+2}}$$

From the 2nd-order Taylor expansion:

$$P(y+\Delta y) \approx P(y) + \frac{dP}{dy}\Delta y + \frac{1}{2}\frac{d^2P}{dy^2}(\Delta y)^2$$

The complete price approximation:

$$\frac{\Delta P}{P} \approx -D\,\Delta y + \frac{1}{2}C(\Delta y)^2$$

Why It Always Helps

▲ Yields fall

Price rises MORE than duration predicts.
The ½C·(Δy)² term adds upside.

▼ Yields rise

Price falls LESS than duration predicts.
The ½C·(Δy)² term cushions the loss.

Net result

Convexity always adds value in both directions — it is an unconditional benefit.

Investors Pay a Premium for Convexity

What drives it higher, how it aggregates, and why callable bonds prove the market prices it.

What Increases Convexity?

Longer Maturity

t(t+1) grows rapidly- distant cash flows dominate curvature. A 30-yr bond is far more convex than a 2-yr.

Lower Coupon Rate

More value sits in the final payment (far in future). Zero-coupon bond = maximum convexity for a given maturity.

Lower Yield Level

Heavy discounting compresses distant cash flow weights. Convexity rises as yields fall- inverse relationship.

Aggregation Formulas

Zero-Coupon

$$C_Z = (T - t)^2$$

Coupon Bond

$$C_C = \sum w_i T_i^2$$

Portfolio

$$C_W = \sum w_i C_i$$

Callable Bonds

Negative Convexity

When rates fall, the issuer exercises the call option and redeems at par-capping the price upside that positive convexity would have delivered.

The bondholder loses the asymmetric benefit at exactly the moment it would be most valuable.

Convexity Premium in Practice

Bond Type	Convexity	Yield
Non-callable	Positive ✓	Lower
Callable	Negative ×	Higher

The yield spread between callable and non-callable = the convexity premium.

Immunization Using Duration and Convexity

What ? Immunization is the strategy of constructing a **portfolio whose value does not change when interest rates move**.

Immunization using (only) duration

Build a portfolio wit 1 unit of a bond and K units of another bond such that:

dW = 0

1. **Value** of our **portfolio**:

$$W = P_1 + k \times P_2$$

2. Portfolio changes in **duration**:

$$dW = -D_1 \times P_1 \times dr + k \times (-D_2 \times P_2 \times dr)$$

To obtain **k**, we use the formula :

$$k = -\frac{D_1 P_1}{D_2 P_2}$$

Example and Limits

Consider the following example:

Long → 10-Y ; 5% Coupon bond

- Dc = 8.0075
- Pc = €104.1413
- Interest rate : increase

Hedge → 10-Y Zero-coupon bond

- Dz = 10
- Pz = €63.6445

$$k = -\frac{D_c \times P_c}{D_z \times P_z} = -\frac{8.0075 \times 104.1413}{10 \times 63.6445} = -1.3103$$

The limit:

Spot Curve Shift	P_c	P_z	Position	dW
Initial Value	104.1413	83.3907	-1.3103	
$dr = 0.1\%$	103.3296	82.5796	-1.3103	-0.0007
$dr = 1\%$	96.3582	75.6418	-1.3103	-0.0343
$dr = 2\%$	89.2724	68.6454	-1.3103	-0.1237
$dr = -0.1\%$	104.9609	84.2102	-1.3103	0.0000
$dr = -1\%$	112.6974	91.9774	-1.3103	-0.0306
$dr = -2\%$	122.1108	101.4972	-1.3103	-0.1370

Immunization using duration and convexity

Re-writing **portfolio change**:

$$\begin{aligned} dW &= -D_c \cdot P_c \cdot dr + \frac{1}{2} \cdot C_c \cdot P_c \cdot dr^2 + \\ &\quad + k \cdot \left(-D_z \cdot P_z \cdot dr + \frac{1}{2} \cdot C_z \cdot P_z \cdot dr^2 \right) \end{aligned}$$

Collecting terms in dr and dr^2, we obtain:

$$dW = -\overbrace{(D_c \cdot P_c + k \cdot D_z \cdot P_z)}^{\text{Duration}} \cdot dr + \frac{1}{2} \overbrace{(P_c \cdot C_c + k \cdot P_z \cdot C_z)}^{\text{Convexity}} \cdot dr^2$$

Hedging both **duration** and **convexity**:

1. **Value** of the **hedged** portfolio:

$$W = P_c + k_s \cdot P_{z,s} + k_l \cdot P_{z,l}$$

2. **Change** in **portfolio value** for shift in i

$$\begin{aligned} dW &= -\overbrace{(D_c \cdot P_c + k_s \cdot D_{z,s} \cdot P_{z,s} + k_l \cdot D_{z,l} \cdot P_{z,l})}^{\text{Duration}} \cdot dr + \\ &\quad + \frac{1}{2} \underbrace{(C_c \cdot P_c + k_s \cdot C_{z,s} \cdot P_{z,s} + k_l \cdot C_{z,l} \cdot P_{z,l})}_{\text{Convexity}} \cdot dr^2 \end{aligned}$$

In Practice: Butterfly Strategy

1. Motivation:

Interest rate changes affect bond prices through duration. To protect a portfolio from small rate changes, we can construct a duration-neutral position.

Exercise VI of our assignment prompts us to carry out the following strategy, targeting a €15m investment:

- Long short maturity (3Y)
- Short medium maturity (5Y)
- Long long maturity (10Y)

The resulting portfolio hedges against small parallel shifts in the yield curve and protects against larger yield moves.

2. Bond Pricing and Risk Measures:

Because bonds trade between coupon dates, we use the dirty price, which reflects the true market value of the bond.

$Dirty\ Price$ = Clean price + accrued interest.

$Dmod$ = First-order sensitivity of price to yield changes.

$Convexity$ = Measures the curvature of the price-yield relationship

$DV01i = \frac{Dmod \cdot Price}{10'000}$ = Price change for a 1 basis point change in yield.

3. Key Conditions and Wing Balance:

Duration neutrality requires: Portfolio DV01 = 0, in casu:

$$k_3 \cdot DV01_3 - k_5 \cdot DV01_5 + k_{10} \cdot DV01_{10} = 0$$

Where:

- $DV01i$ = price sensitivity of bond i to a 1 bp yield change.
- k_3, k_5, k_{10} = normalized hedge ratios.

This ensures first-order effect of rate changes cancels out.

Then, to form the butterfly shape, the two wings must contribute equally to duration:

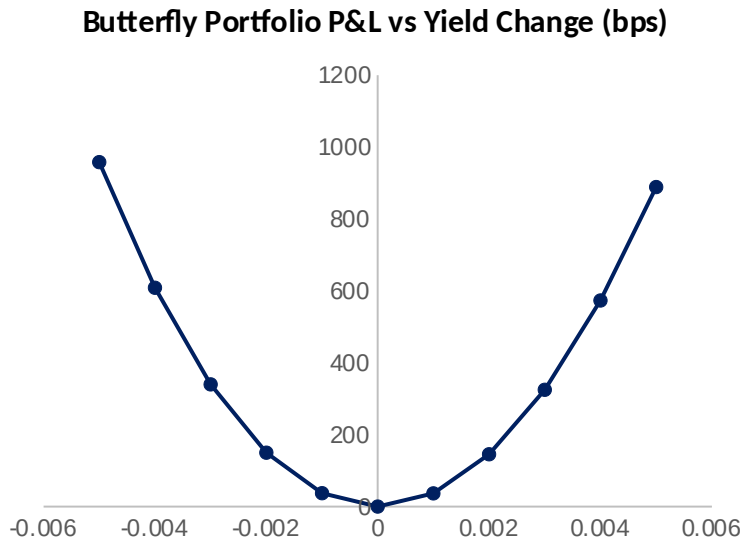
$$k_3 \cdot DV01_3 = k_{10} \cdot DV01_{10}$$

4. Final Portfolio Positions:

Scaling the normalized hedge ratios into actual bond positions so that the total investment equals €15m, while preserving the duration-neutral butterfly condition.

Position	Market Value
Long 3Y	€10.62m
Short 5Y	€14.1m
Long 10Y	€4.38m

$$Scaling\ Factor = \frac{15{,}000{,}000}{(k_3 \cdot P_3 + k_{10} \cdot P_{10})} \cdot 100$$



5. Takeaways:

The butterfly portfolio is duration-neutral (protecting against small rate changes) and positively convex (benefiting from larger yield movements).

The resulting portfolio has a net $DV01 \cong 0$, meaning small rate changes have minimal impact on value, and positive convexity (benefits from larger yield movements):

- Duration-neutrality (risk protection)
- Positive convexity (benefit from curbature)

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Thank You

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